B1.3 Respiration

Summary

Metabolic processes such as respiration are controlled by enzymes. Organic compounds are used as fuels in cellular respiration to allow the other chemical reactions necessary for life.

Underlying knowledge and understanding

Learners should also have some underpinning knowledge of respiration. This should include that respiration involves the breakdown of organic molecules to enable all the other chemical processes necessary for life. Learners should be able to recall the word equation for respiration.

Common misconceptions

Learners commonly hold the misconception that ventilation is respiration. They can also get confused between the terms breakup and breakdown.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
B1.3a describe cellular respiration as a universal chemical process, continuously occurring that supplies ATP in all living cells			WS1.2a	
B1.3b describe cellular respiration as an exothermic reaction			WS1.2b	Demonstration of an exothermic reaction (e.g. heat pack).
B1.3c compare the processes of aerobic respiration and anaerobic respiration	in plants/fungi and animals the different conditions, substrates, products and relative yields of ATP		WS2a, WS2b, WS2c, WS2d	Research into whether plants respire. (PAG B2, PAG B4, PAG B5, PAG B6) Investigation of fermentation in fungi. (PAG B2, PAG B4, PAG B5, PAG B6) Investigation of respiration in yeast using alginate beads to immobilise the fungus. (PAG B2, PAG B4, PAG B5, PAG B6)

B1.3d	explain the importance of sugars in the synthesis and breakdown of carbohydrates	use of the terms monomer and polymer			Demonstration of the synthesis and breakdown of biological molecules (e.g. using Lego bricks). Qualitative testing of biological molecules PAG B2
B1.3e	explain the importance of amino acids in the synthesis and breakdown of proteins	use of the terms monomer and polymer			Qualitative testing of biological molecules PAG B2
B1.3f	explain the importance of fatty acids and glycerol in the synthesis and breakdown of lipids				Qualitative testing of biological molecules PAG B2